

# Machine Learning and Its Objects

Towards Computing Musical Genre

Owen Green

# Preamble

- Continuing thread of disciplinarity
- My focus is on ML as the technical component of 'AI'
- And on modelling for analysis, rather than applications
- Concretely:
  - Using musical genre as a lens, what can we learn about the epistemological horizons of machine learning as practiced in music informatics research (MIR)?
- Tracks my paper, but as a way to think more generally about modelling, research, discipline

# Sonic-Social Genre

“A core topic of music informatics is music ‘similarity’, which is often approached by the proxy task of music genre classification. This proxy task, however, lacks an adequate theory of genre, as well as meaningful experimental methods for evaluating music listening algorithms.”

“The aim is to examine how the nature of genre is and is not compatible with computational methods.”

# A CRITICAL SURVEY OF RESEARCH IN MUSIC GENRE RECOGNITION

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## ABSTRACT

This paper surveys 560 publications about music genre recognition (MGR) published between 2013–2022, complementing the comprehensive survey of [474], which covered the time frame 1995–2012 (467 publications). For each publication we determine its main functions: a review of research, a contribution to evaluation methodology, or an experimental work. For each experimental work we note the data, experimental approach, and figure of merit it applies. We also note the extents to which any publication engages with work critical of MGR as a research problem, as well as genre theory. Our bibliographic analysis shows for MGR research: 1) it typically does not meaningfully engage with any critique of itself; and 2) it typically does not meaningfully engage with work in genre theory.

exhaustively surveying research published between 2013–2022 related to MGR—as well as any earlier publications discovered to have been missed in Sturm’s original work—such that the two surveys give a comprehensive account up to 2022. It also aims to answer several critical questions. Is the experimental design *Classify* and figure of merit accuracy still the most frequent, despite their noted serious flaws threatening the validity of conclusions drawn from them [470, 471, 561, 562]. Is GTZAN [563] still the most used public dataset, despite its noted faults [400, 467, 471, 564]? How have these faults been considered or even reconsidered in the past decade? How has all this new research in MGR engaged with work that is critical of MGR as a research problem, i.e., [231, 232, 329, 401, 467, 469–471, 473, 475–478, 561, 562, 564–574]? How has all this work engaged with genre theory such as [575–598]?

The next section describes the methodology we use to collect and catalogue publications. Section 2 presents our

# Modelling, Research, Discipline

- Why do we build (computational) models?
  - E.g. Explaining; describing; predicting
- How do the (implicit!) intellectual commitments of our disciplines influence this?
- These Qs are key to any effort at reflexive, critical interdisciplinary work
  - But hard!

# Predicting vs Describing vs Explaining

- Prediction *isn't* just a technical property of a model, it's also a matter of disposition
  - Consider linear regression...
- A predictive model requires fewer assumptions
  - So is a good fit for neural network based ML – but not inevitable
- So: we have to be mindful that modelling approaches are partly a matter of technics but also of disciplinary culture
- (Aside: this also goes for 'pure' epistemological categories like induction, deduction etc.)

# So, for genre

- MIR research is **application orientated**
  - “data driven” rather than “knowledge driven” (but which data??)
  - Tends towards modelling for prediction, specifically classification
  - Musical example in, genre label out
- Popular Music Studies more interested in explaining, describing
  - E.g. emergence, evolution, interrelations of particular genres
  - Combines analysis, theory, fieldwork, archival research, and more

# Complex Interdisciplinary Picture

- This isn't just a humanities & social sciences vs STEM fissure
- Consider:
  - Humanities vs social sciences for music already quite fractious
    - Straight musicology vs ethnomusicology & popular music studies
  - Difficult demarcation between science and engineering
    - Outcome Reasoning (Baiocchi and Rodu 2021; 2023): “It works” as criterion for epistemic success
    - Framing measurement problems as prediction problems (Mussnug 2022)



# Some Genre Theory

Really Not Very Much Though

BRACKETT

**Forthcoming, Summer 2027!**

Born and Brackett (eds.) *Music and Genre: New Directions*, Duke

Includes

Green and Born “What’s (In) a Genre? Sounds, Socialities and Time in Early Jungle Drum & Bass”

GENRE AND  
TWENTIETH-CENTURY  
POPULAR MUSIC

Article

## Music and the materialization of identities

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Music is just one of the ways in which we materialize the materialization of identity because it opens up new perspectives on issues of materiality, mediation and affect. These perspectives are intimately related in turn to music's plural socialities, which necessitate a novel approach to theorizing the social. Music, it is proposed, demands an analytics that encompasses four planes of social mediation; while these socialities, with other forms of music's mediation, together produce a constellation of mediations – an assemblage. All four planes of social mediation enter into the musical assemblage: the first two amount to socialities engendered by musical practice and experience; the last two amount to social and institutional conditions that afford certain kinds of musical practice. The four are irreducible to one another and are articulated in contingent ways through relations of synergy, affordance, conditioning or causality. By adopting the topological metaphor of the plane to stand for distinctive socialities mediated by music, the intention is to highlight both their autonomy and their mutual interference. The second half turns to genre theory to suggest that analysing genre in terms of the mutual mediation between two self-organizing historical entities illuminates how social identity formations may be refracted through music and how musical genres can be understood as social formations that have the capacity to create agglomerations of the affective, the affective conditions and consequences of theories of affect, association and entrainment. While such theories illuminate the generative nature of the mutual mediation between musical formations and social formations, they are limited by lack of awareness of the four distinctive planes of music's social mediation, as well as the significance of their autonomy and their contingent interrelations for understanding how music materializes identities.

### Keywords

affect, assemblage, genre, materiality, music, social mediation

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**SAGE**

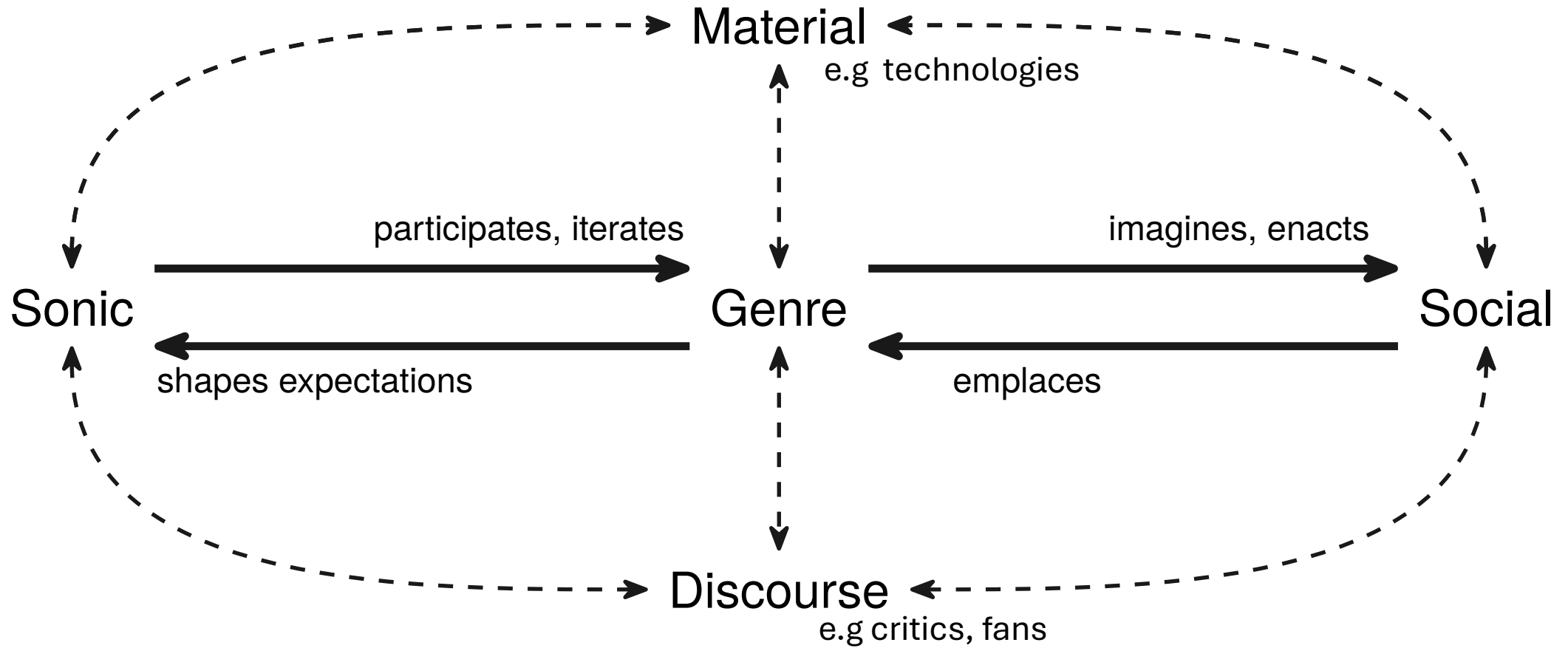
# Genre: “A Motile Assemblage”

- Mutual **mediations** between the social, the sonic, the material
- Genres mutate over time. They also *make* cultural time
  - Genres as *processes*
- *Works* (tracks, texts...) *participate* in genres, are not “members”
- Genres can be highly perspectival...
- Destabilises any idea of ground truth

# Mediation

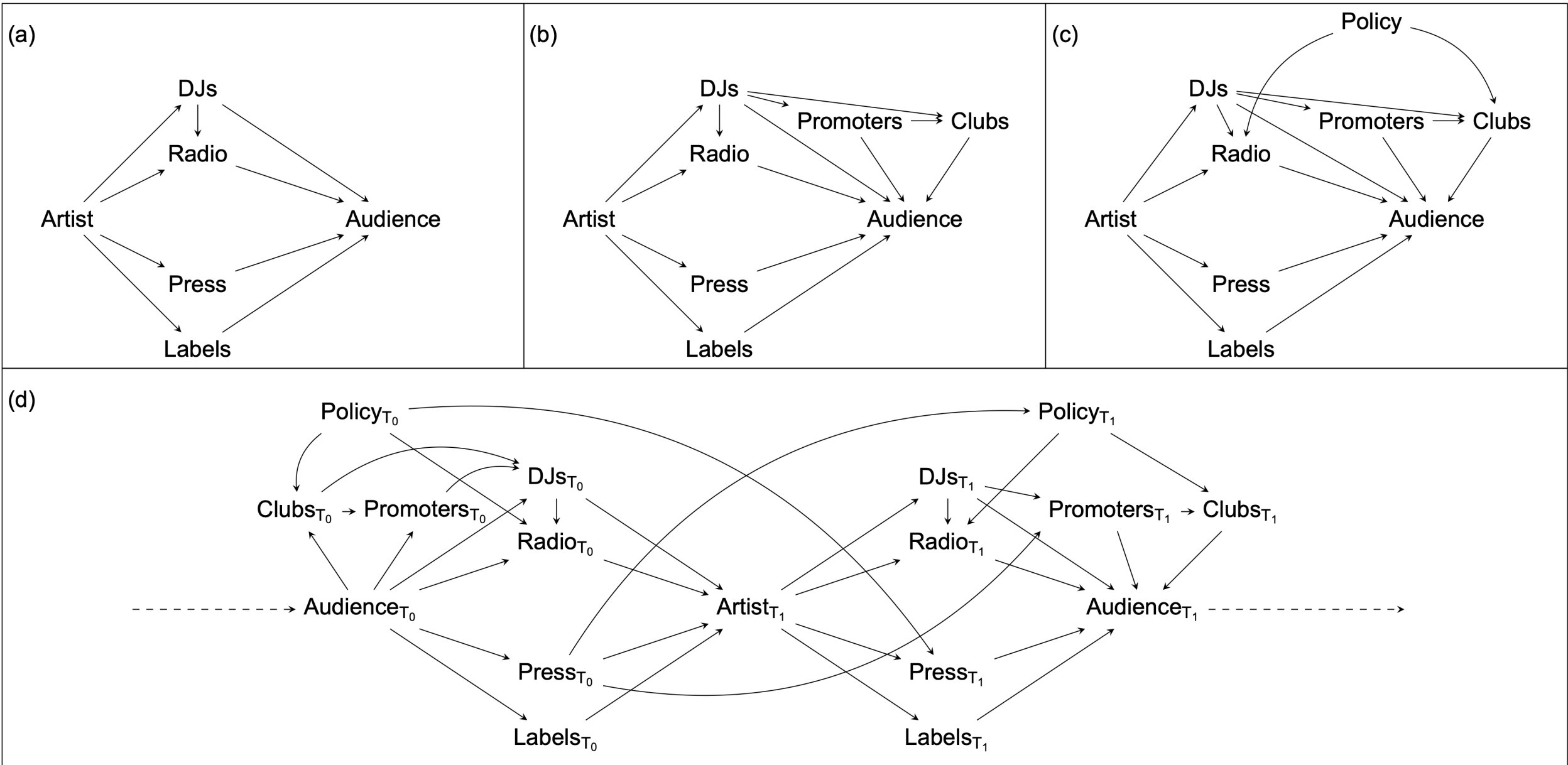
- Is musical experience *immediate*?
  - It is not!
  - It is *mediated* by the sonic, social, cultural, technological etc
  - See Born 2005, 2010, 2011
- Is *social* experience immediate?
  - Also, no
  - Music is one of the things that mediates it
- Hence *mutual mediations*

# Mutual Mediations



# For example, in Jungle Drum n' Bass

- Crucial to grasp social mediations
  - Racial politics
  - Genre produces socialities, e.g. as multi-racial youth scene
  - Role of institutions: broadcasters, press, state, labels, promoters
- Material mediations:
  - Samplers, sound-systems, radio transmitters
- Time
  - Rapid production of sub-genres
  - Reflects *and drives* mutability of other mediations



# In Sum

- To analyse a genre along these lines means tracing the “motility” of an evolving cast of mediations
- Figuring out which are important and when is hard
- Implies great diversity of data modalities and sources
- Also implies tight and evolving link between what we think genre or a particular genre *is* and what we know about it
  - For Born (2010), “analytical ontology”
  - cf. Lynch (2013), “ontography”



# Existing Computational Approaches to Genre

# Genre in Music Informatics

- Genre as
  - musical similarity
  - ‘high level’ feature
- Genres as *sets*
- A few shared datasets
  - Ever larger, but not ever richer
- Committed to being application focused, data-driven
- Outcome reasoning
- Epistemological object is algorithms, not genre
- Frequently acknowledged as difficult (‘fuzzy’, ‘subjective’)
  - But few moves to strengthen relationship to theory

# Limits are widely acknowledged

“Musical genres have no strict definitions and boundaries as they arise through a complex interaction between the public, marketing, historical, and cultural factors... However even with current musical genres, it is clear that the members of a particular genre share certain characteristics typically related to the instrumentation, rhythmic structure, and pitch content of the music.”

(Tzanetakis and Cook 2002, 1)

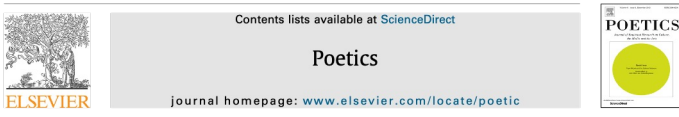
- But are they *understood*?
  - See Salamon (2019) → Identifies important limits, not where to look for possible answers!

# Epistemology

- Predictive thrust
  - Influence of Data Science + Breiman's 'two cultures of statistics' (?)
- Proceeds by improving benchmark evaluations against (public) datasets
  - Remember 'outcome reasoning'
- Nomothetic by inclination
- Ambivalent re: theory
  - Results tell us more about computers than about music

# Other Computational Approaches

- Small amount of work on *musical* genre in Computational Social Science; Cultural Evolution
  - Interestingly, *repeats* “outcome reasoning”, this time bracketing out the sonic at the expense of (proxies for) the social
- More sustained work on literary genre in Computational Literary Studies
  - Imaginative uses of predictive models, but...
  - Reveals gulf between humanities and social sciences in concerns with provenance of data
  - Underscores major difference between text and music: *no ‘natural’ segmentation of music*



## A community ecology of genres Explaining the emergence of new genres in the UK field of electronic/dance music, 1985–1999

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### ABSTRACT

This paper proposes and tests a 'cultural ecological' model for explaining the emergence of 12 genres in the UK field of electronic/dance music. By combining two large data sets, containing more than eight thousand documents from the mainstream UK music press and discographical information on almost fifty thousand records released by UK labels between 1985 and 1999, this study finds evidence that genre emergence is affected by ecological mechanisms of cross-form legitimation and competition. By taking into account cultural similarities between genres, 'density' of production in each genre, and symbolic legitimation of genres by the music press, it is possible to predict when and whether new genres emerge. The legitimation of a genre identity – signaled both by increasing density and sudden increases in media attention – invites the emergence of new genres of a similar identity. Genre identity positions, however, also reach a saturation point, at which time the identity position no longer leads to the founding of new genres.

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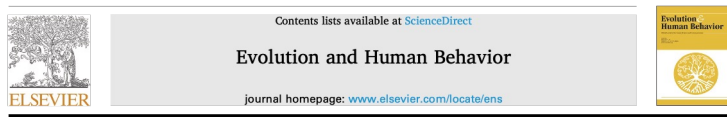
### 1. Introduction

Sociologists of art and culture have often studied the problem of why and when new categories, i.e. genres, emerge in cultural fields (Baumann, 2007a; Bourdieu, 1996; DiMaggio, 1982; Dowd, 2003; Ennis, 1992; Lopes, 2002; Peterson, 1990, 1997; Roy, 2010; Santoro, 2002; White & White, 1965). Although these studies have provided much insight into the conditions and factors that explain the emergence of new art forms, it is, however, notable that these studies predominantly consist of historical, qualitative case studies which select a particular instance of form emergence – the rise of the novel, impressionism, rock 'n' roll – and reconstruct the social and institutional factors that contributed to its emergence. Much less common are analyses of more general, law-like regularities across multiple cases of form emergence, rather than *within* case analysis (cf. Goertz & Mahoney, 2012). Although scholars have proposed theoretical generalizations by comparing across cases of art world development (Baumann, 2007b; Lena, 2012), empirical studies of multiple cases attempting to test general mechanisms of form emergence have been scarce.

The goal of this paper is to contribute to the study of these more general mechanisms of genre emergence. To do this, I borrow theoretical and methodological insights from the sociology of organizations, more particularly 'organizational ecology' and the study of the emergence of organizational forms. Although in the past organizational ecologists defined organizational forms in realist ways and considered the delineation of organizational populations as relatively

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## Phylogenetic reconstruction of the cultural evolution of electronic music via dynamic community detection (1975–1999)

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### ABSTRACT

Phylogenetic trees or networks representing cultural evolution are typically built using methods from biology that use similarities and differences in cultural traits to infer the historical relationships between the populations that produced them. While these methods have yielded important insights, researchers continue to debate the extent to which cultural phylogenies are tree-like or reticulated due to high levels of horizontal transmission. In this study, we propose a novel method for phylogenetic reconstruction using dynamic community detection that focuses not on the cultural traits themselves (e.g., musical features), but the people creating them (musicians). We used data from 1,498,483 collaborative relationships between electronic music artists to construct a cultural phylogeny based on observed population structure. The results suggest that, although vertical transmission appears to be dominant, the potential for horizontal transmission (indexed by between-population linkage) is relatively high and populations never become fully isolated from one another. In addition, we found evidence that electronic music diversity has increased between 1975 and 1999. The method used in this study is available as a new R package called *DynCommPhylo*. Future studies should apply this method to other cultural systems such as academic publishing and film, as well as biological systems where high resolution reproductive data is available, and develop formal inferential models to assess how levels of reticulation in evolution vary across domains.

### 1. Introduction

Historically, researchers have relied on phylogenetic comparative methods from biology to create phylogenetic trees or networks of cultural evolution known as "cultural phylogenies". These methods use differences and similarities in the cultural products of different populations (analogous to differences and similarities in DNA) to reconstruct the historical relationships between them. Traditional phylogenetic methods, which assume the tree-like structure typical of genetic evolution, have yielded critical insights, particularly in linguistics (Bouckaert et al., 2012; Levinson & Gray, 2012; Pagel, Atkinson, & Meade, 2007). However, researchers have debated whether cultural phylogenies are fundamentally tree-like, or whether high levels of horizontal transmission lead to a more reticulated structure (Borgerhoff Mulder, Nunn, & Towner, 2006; Boyd, Borgerhoff-Mulder, Durham, & Richerson, 1997; Cabrera, 2017; Gray, Bryant, & Greenhill, 2010;

Rivero, 2016). For example, biologist Stephen Jay Gould wrote that:

"Biological evolution is a bad analogue for cultural change [...] Biological evolution is a system of constant divergence without subsequent joining of branches. Lineages, once distinct, are separate forever. In human history, transmission across lineages is, perhaps, the major source of cultural change." – Gould (1991).

Certainly, this is an oversimplification. Horizontal transmission also frequently occurs in biology and sometimes provides problems for the practice of classification (Doolittle, 2009). Horizontal transmission is for instance the primary mechanism for the spread of antibiotic resistance in bacteria (Gyles & Boerlin, 2014), but it also occurs in vertebrates, invertebrates, and plants (Bock, 2010; Crisp, Boschetti, Perry, Tunnacliffe, & Micklem, 2015). Nevertheless, Gould had a fair point. Although cultural traits evolve through a process of variation, selection and

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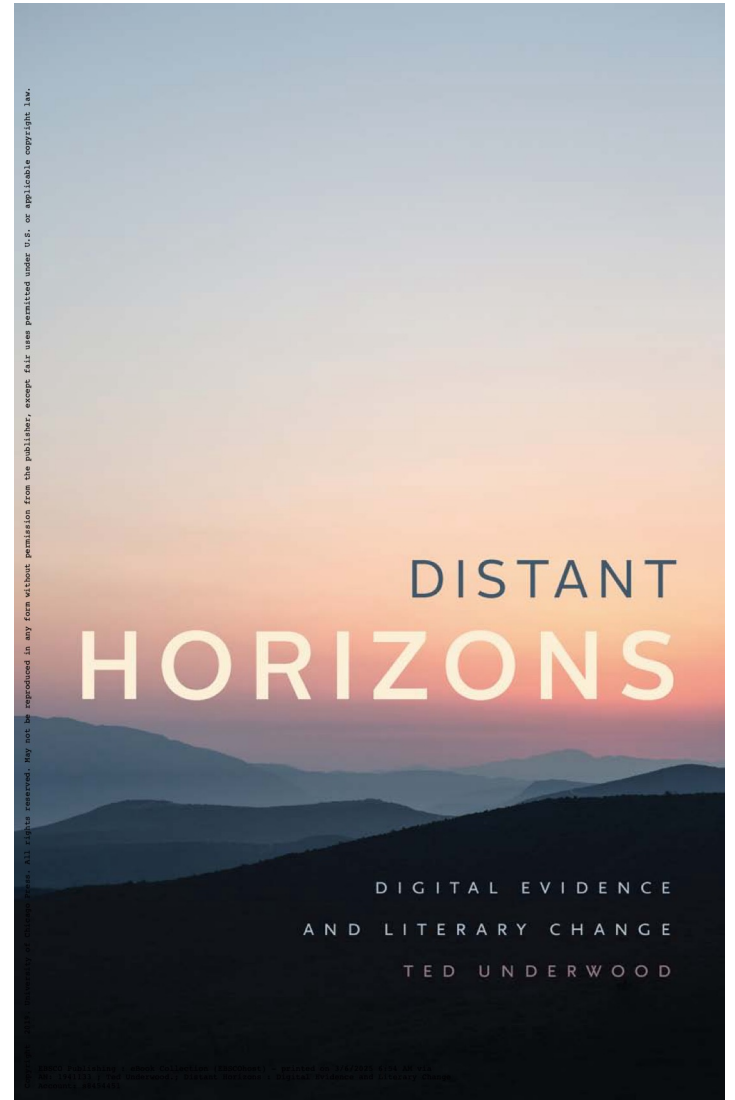
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# Hot Takes

- MGR would not form basis of computational approach to GT
- Just doing different things! (Explaining vs predicting)
  - Notwithstanding clever things done with prediction by Ted Underwood
- *However*, barriers to a more 'relational' MIR not primarily technical or due to incommensurability of quant vs qual (and so on)
- More a matter of institutional conditions, disciplinary dispositions
- IMNSHO: Better interdisciplinarity → Better science
  - Hampered by mutual incomprehension, lack of shared concepts

# Mutual Mediations of Models, Objects, Data

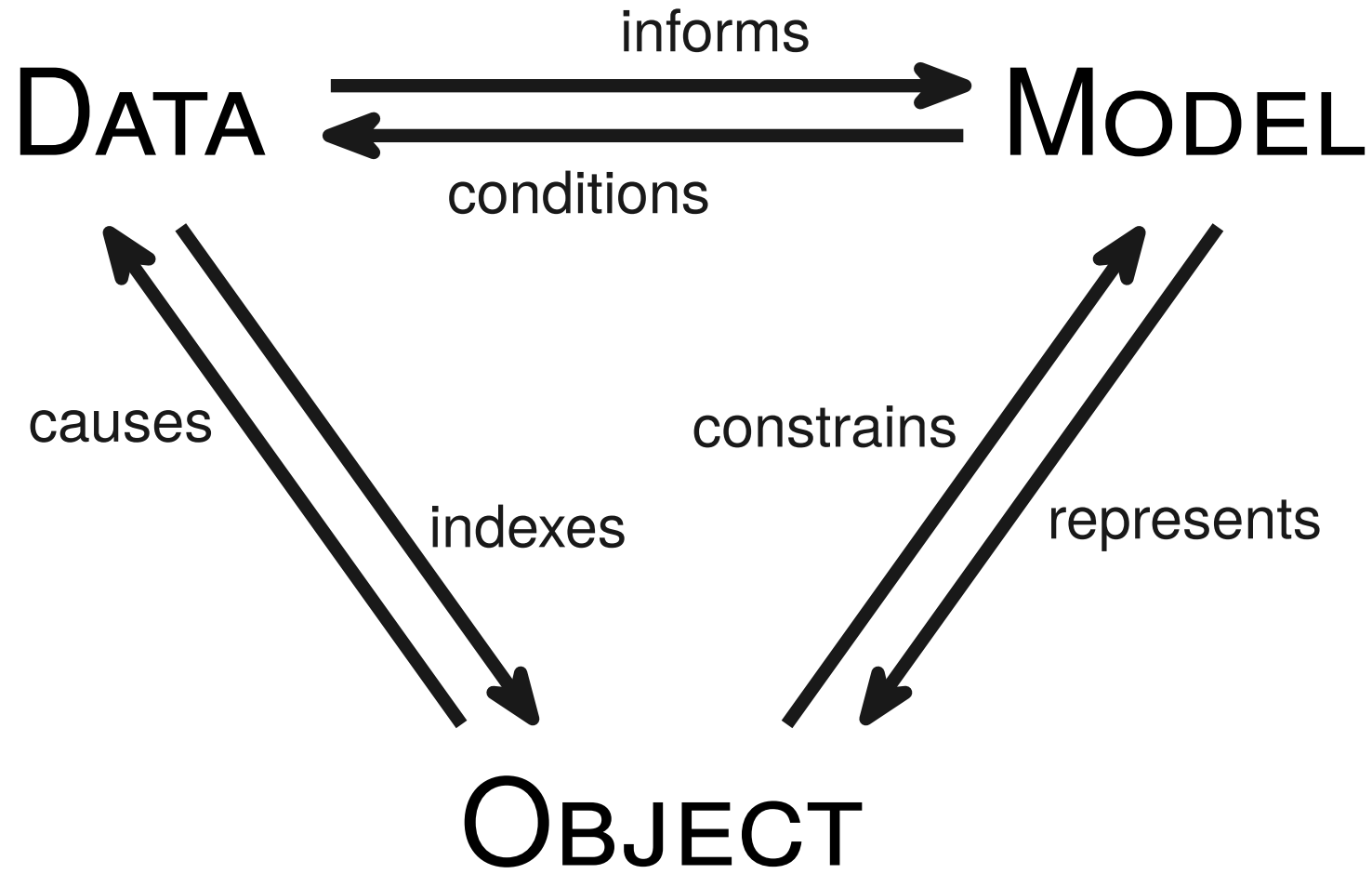


Fig 5 in Kockelman (2020)



# Topics for Discussion

- **Models:** What is modelling typically used *for* in your discipline(s)? What about in your personal research?
- **Objects:** How much of what you know about your musical objects is informed by models; how much are your models informed by what you know?
- **Data:** Where do your data come from? Do they index what you *actually want to measure*, or something else? Is more data always good (for your research)?